STRUCTOR SOLUTIO

Technology in Business

Why digital fluency is no longer optional — Instructor Solution

Gerrish School of Business · Endicott College · Fall 2026

Connect to Prior Knowledge

In Lessons 01 and 02, you set up your professional digital workspace and learned how BUS123 is structured. Now we zoom out to answer a bigger question: *why does any of this matter?* Technology is not a background detail of modern business — it is the infrastructure that business runs on. This lesson builds the conceptual foundation for everything that follows: understanding where technology fits in professional life, what digital fluency actually means, and why the habits you build now determine how you show up in your first job.

Concept Overview

Technology by the Numbers

A few data points frame the scale of technology in professional environments:

Statistic	Source / Context
1.2 billion Microsoft 365 users monthly	Microsoft — the most widely adopted productivity platform in the world
80% of Fortune 500 run on Microsoft 365	Microsoft enterprise data — the standard for large-organization professional tools
67% of jobs require advanced digital skills	McKinsey Global Institute — beyond basic email and document use

Passive User vs. Professional Operator

There is a meaningful difference between using technology and operating it professionally. Most people are passive users — they use tools as given, with default settings, and look up help every time something goes wrong. A professional operator builds systems, writes formulas, structures data for others, and solves novel problems without guidance.

Passive User	Professional Operator
Types answers into cells	Builds formulas with cell references

Uses default settings	Customizes tools for the task
Accepts data at face value	Questions data sources and accuracy
Searches for help every time	Diagnoses and resolves errors independently
Does the task once manually	Automates repeatable work

Data Fluency: Read, Question, Act

Data fluency is often misunderstood as a purely technical skill. In reality, it is the ability to work with data meaningfully — understanding what it shows, challenging its assumptions, and using it to support decisions. The three-step framework:

Step	What It Means	Example
Read	Understand what a chart or table is actually showing	Sales are up 12% — but vs. which prior period?
Question	Challenge the source, scope, and accuracy of the data	Are returns included? Which channels? What is excluded?
Act	Use the analysis to make or support a business decision	Recommend increasing inventory for top-selling SKUs

■ Most organizations have more data than they know what to do with. The professionals who advance are not the ones who generate the most data — they are the ones who can tell a clear story from it.

AI Tools and Human Judgment

AI tools are reshaping professional workflows — but they are not replacing the need to understand the underlying work. The key distinction is between what AI does well (drafting, suggesting, summarizing) and what still requires human judgment (verifying correctness, applying context, being accountable for decisions). Professionals who understand their tools deeply will always be more valuable than those who rely on AI to produce outputs they cannot explain.

■ The professional who understands Excel deeply will always be more valuable than one who relies on AI to produce outputs they cannot explain. Build the skill first — then use AI to accelerate it.

Key Reference — Technology by Industry

Industry	Case Study Company	Key Technology Uses
Retail	Tidal Goods Co.	POS data, inventory tracking, demand forecasting, markdown analysis
Financial Services	Meridian Advisory Group	Payroll systems, TVM models, loan amortization, tax calculation
Event Planning	Anchor & Oak Events	Breakeven models, cash flow projections, labor scheduling
Healthcare	Harborside Medical Center	Payer mix dashboards, patient data analysis, complex payroll

Check Your Understanding

Answer these 7 questions after reading. Answers appear at the end of this document.

1. Explain the difference between a passive technology user and a professional technology operator. Give one specific example of each behavior in an Excel context.

2. A classmate says: "I already know Excel — I use it for my personal budget." Based on the fluency spectrum, what level does this likely represent, and what would it take to reach Professional Operator level?

3. Choose one of the four BUS123 case study companies and describe two specific ways technology supports its operations. Be specific — name the type of tool or analysis used.

4. Define data fluency in your own words. Then give an example of how the "Question" step of data fluency (read, question, act) would apply to a sales report.

5. What is one skill that AI tools currently do well, and one task where human judgment is still required? Why does the distinction matter for your career?

6. You are interviewing for a marketing coordinator role. The job description mentions Excel and data reporting. Describe two specific BUS123 skills that directly apply to this role.

7. What does it mean for technology to be "the infrastructure business runs on"? Use one statistic from this lesson and one example from a real industry to support your answer.

Key Vocabulary

Digital Fluency

The ability to use technology tools purposefully and professionally — not just operating them, but adapting them to solve specific problems, evaluating their outputs, and integrating them into workflows.

Data Fluency

The ability to read, question, and act on data. Involves understanding what a dataset shows, challenging its assumptions and sources, and using it to support business decisions.

Passive User

Someone who uses technology as given — with default settings, no customization, and without building systems or formulas that scale. Treats results as outputs to type rather than formulas to build.

Professional Operator

A technology user who builds scalable systems, structures data for others, diagnoses and resolves errors independently, and uses tools to solve novel problems without external guidance.

Fluency Spectrum

A five-level framework describing stages of technology proficiency: Basic User, Functional User, Skilled User, Professional Operator, and Expert. BUS123 targets Professional Operator level.

Formula Thinking

The habit of expressing business logic as Excel formulas rather than typed values — so that results update automatically when inputs change and the model remains valid for repeated use.

AI Tool Proficiency

The ability to use AI tools effectively while maintaining judgment about the accuracy and appropriateness of their output. Includes knowing when to trust AI suggestions and when to verify independently.

Answer Key — Check Your Understanding

■ Instructor use only. Do not distribute to students.

1.

A passive user treats Excel like a calculator — typing the final answer into a cell instead of building a formula. They use default features and look up help every time something goes wrong. A professional operator builds formulas with cell references so results update automatically, structures data so others can use it, and diagnoses errors independently. Passive example: typing 247.50 into a result cell. Operator example: writing =B5*C5 so the result recalculates when inputs change.

2.

Using Excel for a personal budget is likely Functional User level — the person uses basic features like SUM and simple tables, but has not built models or diagnosed complex formula errors. To reach Professional Operator level, they would need to write formulas from scratch for unfamiliar problems, use functions like IF, VLOOKUP, and PMT correctly, structure data for others to use, and solve novel problems without guidance or sample files to copy from.

3.

Any two of the following per company: Tidal Goods Co. — POS data analysis to track which products sell fastest, demand forecasting model to decide how much inventory to order. Meridian Advisory Group — payroll calculation system, loan amortization model. Anchor & Oak Events — breakeven spreadsheet, cash flow projection. Harborside Medical Center — payer mix dashboard showing revenue by insurance type, staffing model.

4.

Data fluency is the ability to understand what data is showing, question its source and accuracy, and use it to make or support a business decision. For a sales report, the Question step would involve asking: what time period does this cover, are returns included or excluded, which sales channels are counted, and is the growth percentage calculated from an unusual baseline — before trusting the summary and acting on it.

5.

AI does well at drafting text, suggesting formulas, explaining concepts, and speeding up repetitive research. Human judgment is still required to determine whether the AI output is correct, to apply business context the AI does not have, and to be accountable for final decisions. The distinction matters because professionals who rely on AI without understanding the work cannot explain or defend their outputs — which is a career and ethical liability.

6.

Any two of: Excel for building campaign performance tables and tracking metrics over time (Excel Skills track); data interpretation skills to read and question summary reports (Data Fluency); professional file naming and

OneDrive organization for sharing deliverables with a team (Intro track); formula construction so reports update automatically when data changes (Formula First rule).

7.

When technology is "the infrastructure business runs on," it means every major business function — from paying employees to tracking inventory to serving customers — is built on and delivered through software systems. Supporting statistic: 1.2 billion people use Microsoft 365 monthly, or 80% of Fortune 500 companies run on it. Industry example: a retail company's ability to know which products are selling in real time depends entirely on its POS and inventory management software — without it, buyers would be ordering based on guesses, not data.